PAX alignment run

RUN

155939_pax-live_TEC-active_quick-test
Status: completed

ACQUISITION

Duration: 905.8 s; 6190 fresh samples; 6.90 Hz over recorded samples.

RUN COMMENT

TEC active, setpoint at 10.75KOhms, currently hovering around 10.67kOhms. running for an hour ish.

SETUP

In front of NPBS 1, C port
V_pi expectation only; verify channel and measured driver gain before calibration.

DEFINITIONS

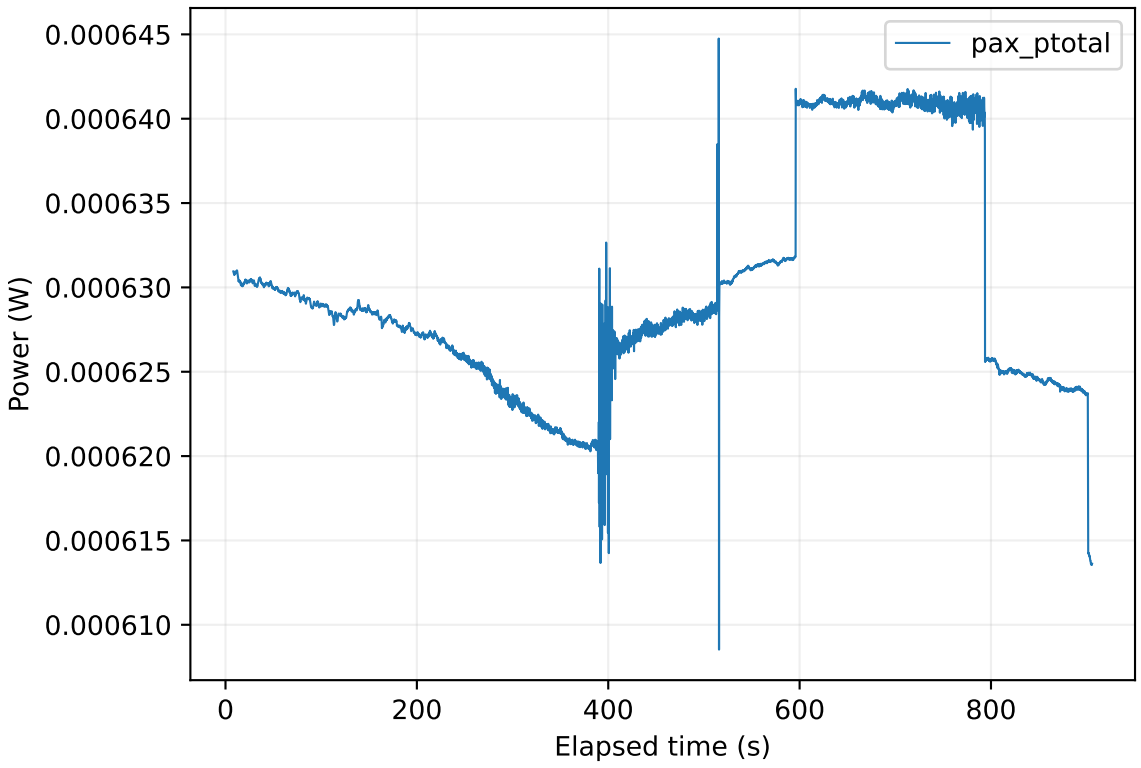
Power: watts. DoP: fraction. S1–S3: the suite's normalized direction (not multiplied by DoP). CSV theta/eta: radians; displayed angles: degrees. No fit or smoothing.

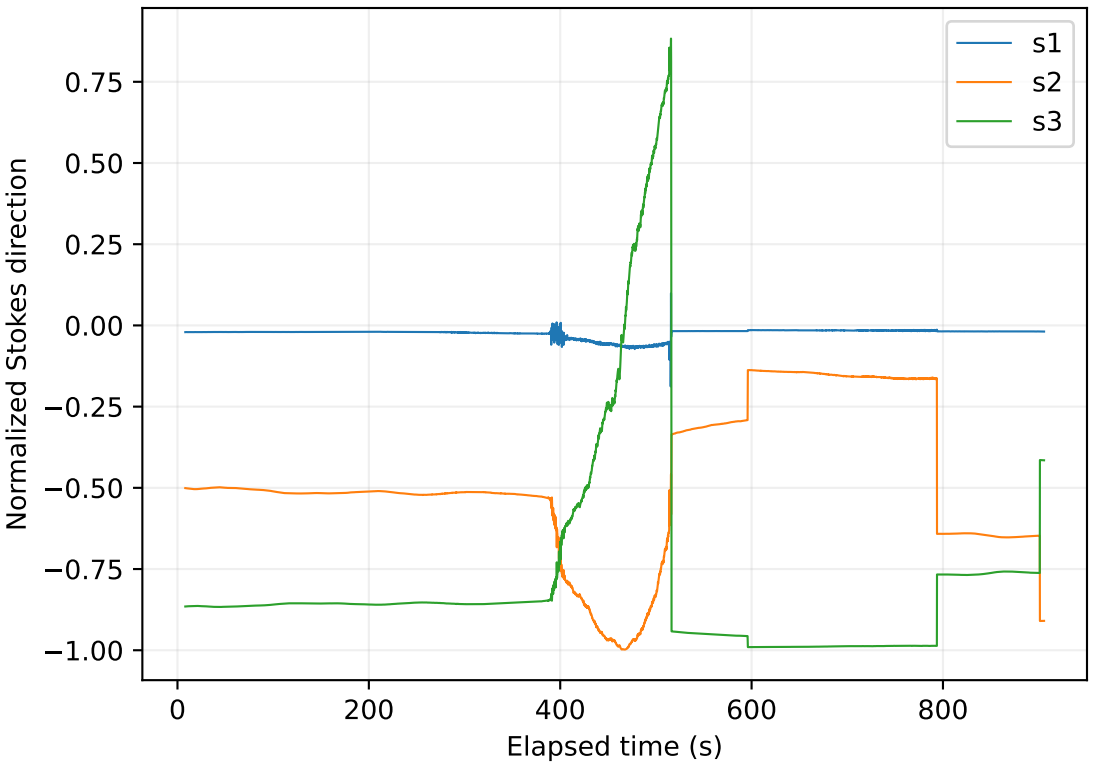
RAW DATA

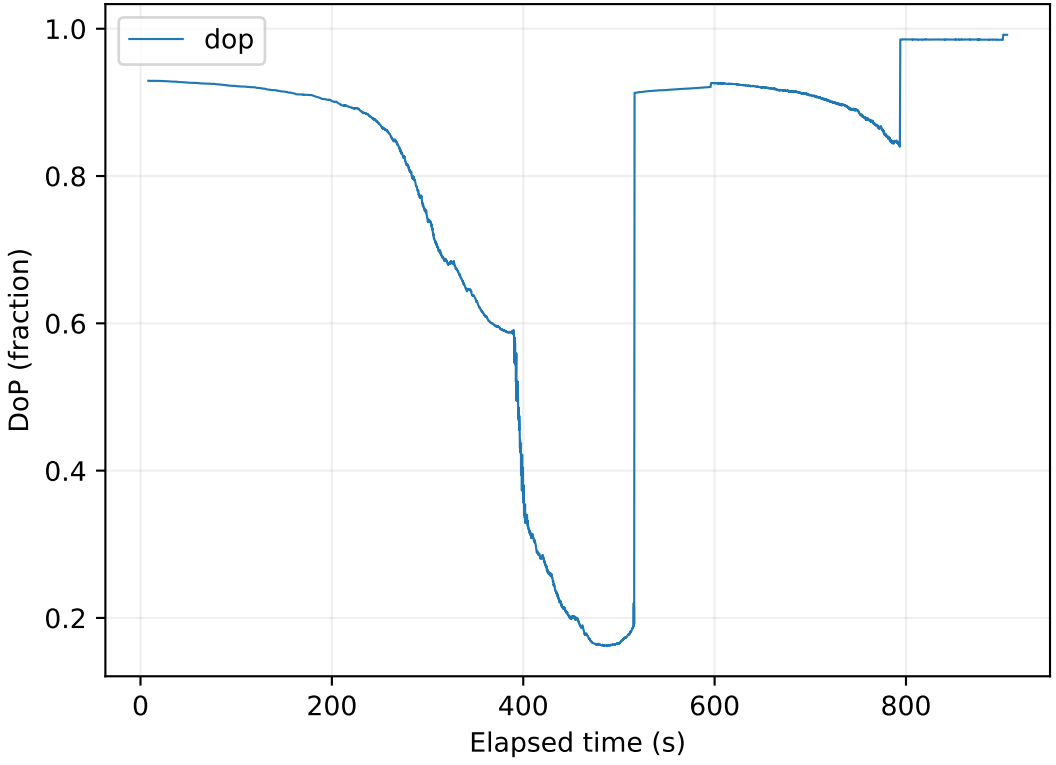
data.csv contains every accepted sample and the full exposed PAX record. Red Pitaya was not connected or measured. Settings/provenance: recipe.json and run.json.

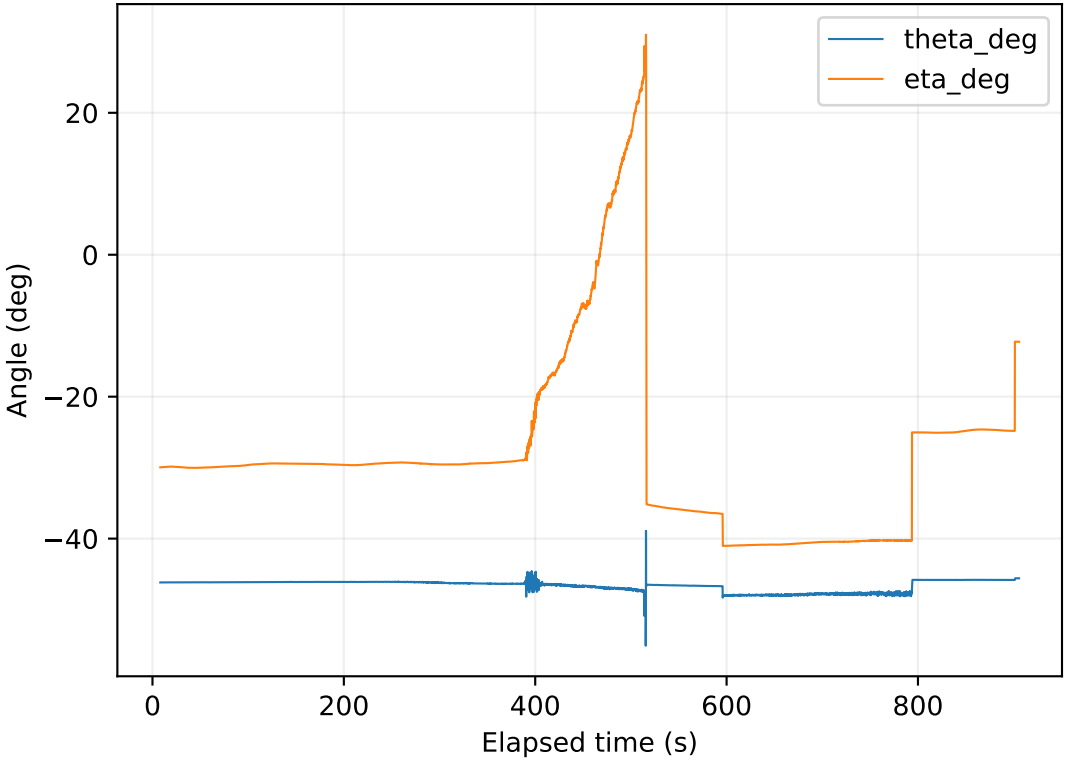
ACQUISITION ISSUES

None

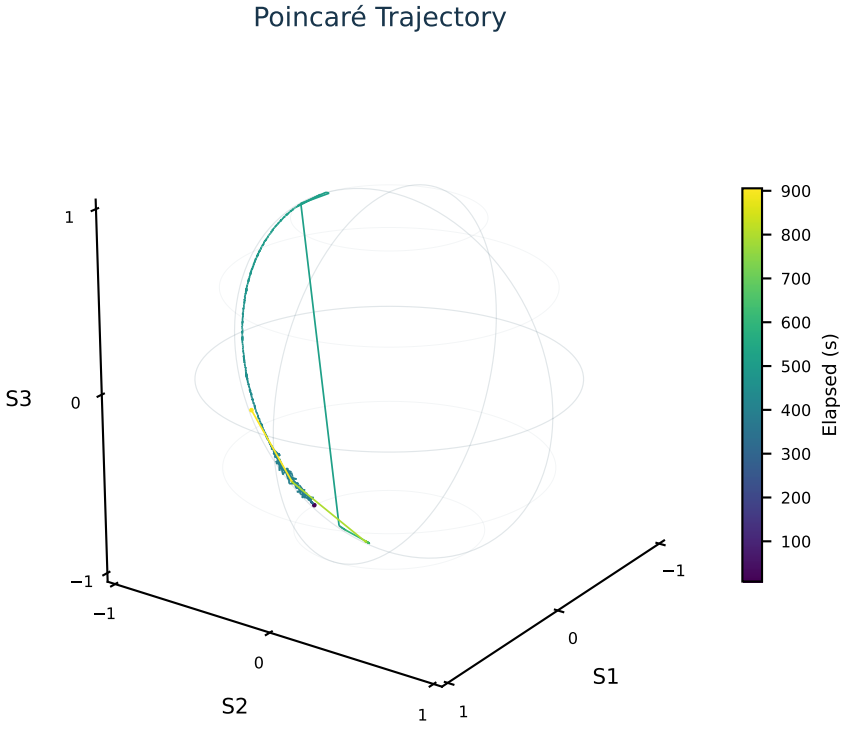

pax_ptotal
Power (W)
0.000645
0.000640
0.000635
0.000630
0.000625
0.000620
0.000615
0.000610
0
200
400
600
800
Elapsed time (s)


s1
s2
s3
Normalized Stokes direction
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
200
400
600
800
Elapsed time (s)


dop
DoP (fraction)
1.0
0.8
0.6
0.4
0.2
0
200
400
600
800
Elapsed time (s)


theta_deg
eta_deg
Angle (deg)
20
0
−20
−40
0
200
400
600
800
Elapsed time (s)

Polarization trajectory and descriptive statistics



Measured normalized Stokes trajectory, colored by elapsed acquisition time. Recorded suite direction (derived from PAX angles; not DoP-scaled). Low-DoP measurements have increased polarization-direction uncertainty; no DoP filtering. Color: recorded host elapsed seconds. No connections over invalid rows, nonincreasing time, or gaps > 0.724 s (5× median positive cadence). UTC comparison tolerance: 0.05 s (max of 0.05 s and 10% median cadence). UTC/elapsed disagreement detected; host elapsed retained and inconsistent adjacent clock intervals unconnected. 0 invalid Stokes/time rows unplottable; finite low-DoP records retained.

Normalized Stokes direction

Dimensionless; variance dimensionless².

Field	N	Mean	SD	Variance	Min	Max
s1	6190	-0.02369	0.013475	0.00018156	-0.18808	0.098973
s2	6190	-0.48172	0.23013	0.052961	-0.99803	-0.13737
s3	6190	-0.77563	0.33567	0.11267	-0.99041	0.88314

Recorded angles

Degrees; variance deg². Linear statistics; no unwrapping.

Field	N	Mean	SD	Variance	Min	Max
theta_deg	6190	-46.626	0.75174	0.56512	-55.092	-38.91
eta_deg	6190	-28.212	12.756	162.71	-41.03	31.012

Statistics describe the entire recorded run, including drift and abrupt jumps; they are not measurement uncertainties or short-term noise estimates. Each finite sample has equal weight (not time weighting). N is the finite count per field out of 6,190 rows; invalid entries are excluded only from that field's statistics. No low-DoP filtering. SD and variance use N−1; unavailable for N<2. Angle means/SD/variance are linear statistics of the displayed wrapped angles; a wrap boundary can inflate spread. No circular-statistics interpretation is implied.